

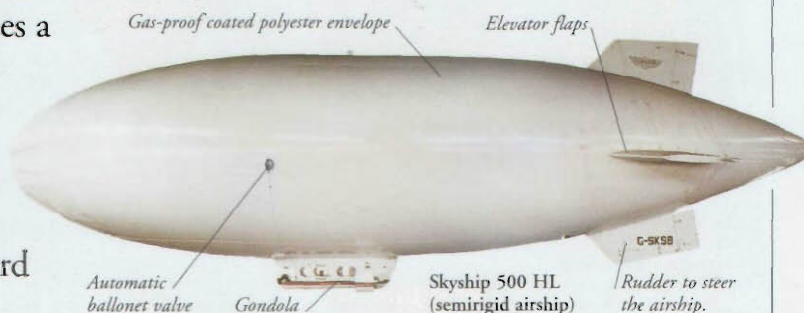
AIRSHIPS AND BALLOONS



AIRSHIPS AND BALLOONS are known as lighter-than-air aircraft because, instead of wings, they use a large envelope, or bag, full of gas or hot air that is lighter than the air in the atmosphere around it. The air pushes the envelope upwards, just as water pushes a submerged air-filled ball upwards. In 1783, the Montgolfier brothers achieved the first manned flight ever by sending a hot-air balloon over Paris. Balloons fly where the wind blows them; airships have engines and can be steered. Today, airships are used for aerial filming and coast-guard patrols, and ballooning is a popular sport.

Anatomy of a modern airship

The main part of an airship is its envelope, which contains bags of helium gas. The gas is slightly pressurized to keep the envelope in shape. A fin and tailplane keep the airship steady as it flies slowly along. The crew travels in a gondola attached to the underside of the envelope.



The Hindenburg, 1937

Airship disasters

Several terrible disasters made people lose trust in airship travel. Airships were usually lost for two reasons: either they were uncontrollable in bad weather; or the highly inflammable hydrogen gas used inside the envelope exploded. Today, airship pilots use the much safer helium gas in special nylon envelopes. However, they still have to be wary of the weather.

Types of airship

Practical airships could be built only after the lightweight internal combustion engine had been developed. The earliest airships were "nonrigid" (they are still used today). These were followed by the "rigid" and the less usual "semirigid" types of airship.



Nonrigid airships have a flexible fabric envelope, from which the load hangs, suspended by ropes.

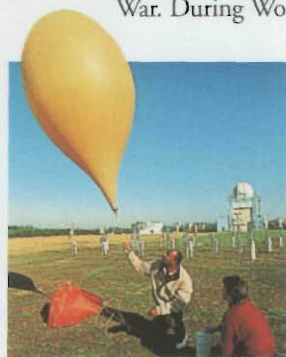


Rigid airship's envelope is built around a rigid framework. This skeleton contains bags of the lifting gas – helium.

Balloons

Balloons were first used for aerial reconnaissance during the French Revolution, and used again in the American Civil War. During World Wars I and II, balloons

were used to spot targets for artillery attacks, and barrage balloons defended cities against aircraft.



Weather and research balloons

To study what is happening in the upper reaches of the atmosphere, pilots send up helium-filled weather balloons. These carry instruments which measure temperature, wind speed, and so on, and send their results to the ground or to satellites by radio.



Balloon festivals

Today, ballooning is a popular sport. During the summer, ballooning enthusiasts gather at festivals to enjoy the dazzling prospect of dozens of brightly coloured balloons flying together. Some of the balloons are owned by companies, and are made in the shapes of their products, as a form of advertising.

Ferdinand von Zeppelin

German count Ferdinand von Zeppelin (1838–1917) began experimenting with air travel in 1891. In 1900, he devised the first airship, a 128-m (420-ft) rigid craft named the LZ1. During World War I, some 100 Zeppelins were built for military use.



Flight

Hot-air ballooning requires a perfectly clear day with a gentle breeze. Too high a wind puts the balloon at risk on take-off and landing. After take-off, a ground crew follows the balloon in a vehicle to recover both it and the crew after landing.



1 The balloon is laid out on the ground. Burners heat air to fill the balloon.



2 The balloon's envelope expands as the hot air starts to fill it



3 The expanding balloon becomes buoyant, and rises into the air.



4 Guy ropes hold the balloon down until the crew boards.



5 The crew blasts hot air into the envelope to keep the balloon afloat.

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